

Comprehensive Analysis and Dietary Strategies with Tableau

A College Food Choices Case Study • Folder 3: Project Design Phase

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Host Organization:	SkillWallet Internship Program
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Folder 3 Metadata & Project Registry

System Registry Parameter	Value Specification
Development Phase	Folder 3 – Project Design (Blueprint Architecture)
Architectural Focus	Relational Schema Modeling, Wireframe Grid Layouts, Tableau Extract Design
Project Status	Validated - Blueprint Ready for Preprocessing Phase

1. System Architecture Design

The project follows a standard star-schema data flow model. Raw questionnaire data is routed through a sanitization pipeline, converted into a locally optimized `.hyper`` extract for fast interactivity, and hosted on the Tableau cloud directory.

2. Entity-Relationship Modeling (ERM)

The schema design maps three core relational nodes for interactive cross-filtering:

Data Node	Primary Attributes
Fact: Behavioral Metrics	Response_ID, Calorie_Intake, Sugar_Level, Fat_Intake, Survey_Date
Dim A: Student Demographics	Student_ID, Gender_Code, GPA_Score_Band, Activity_Frequency
Dim B: Environment Context	Stress_Index, Financial_Budget_Tier, Sleep_Hours, Study_Load

3. Dashboard Interface Wireframe Layout

The visualization workspace uses a partitioned three-zone grid for high-speed data engagement:

- **Zone 1 (Header Panel):** Displays the global KPI Summary including average student cohort health scores and participation density.

- **Zone 2 (Side Filtering Bar):** Provides dynamic multi-select filters for Gender, Financial Status, and Exercise Frequency parameters.
- **Zone 3 (Core Viewport):** Contains the primary Pearson correlation scatter chart (Stress/ Calorie Intake) and comparative analysis bar plots for budget-tier dietary quality.

4. Folder 3 Design Conclusion

This project blueprint establishes a clear architectural design path for the analytical workspace. By finalizing the ERM relations and user wireframe layouts, the foundation is secured to proceed directly into the Data Preprocessing and Workbook Construction phases.

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Governance Review:

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